



PeerPods

A Context-Aware UI for Music
Reflection and Community



Welcome to PeerPods

Ena Helena Otuokon
Student ID: 24013854

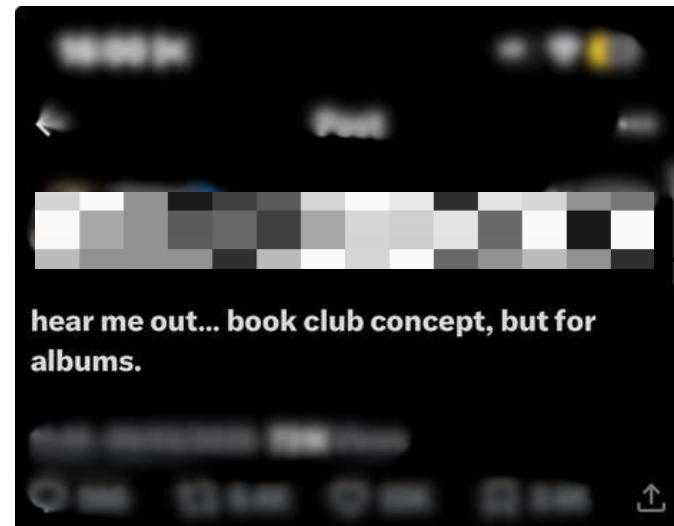
Module: KV7005 Intelligent User
Interfaces

PROBLEM STATEMENT

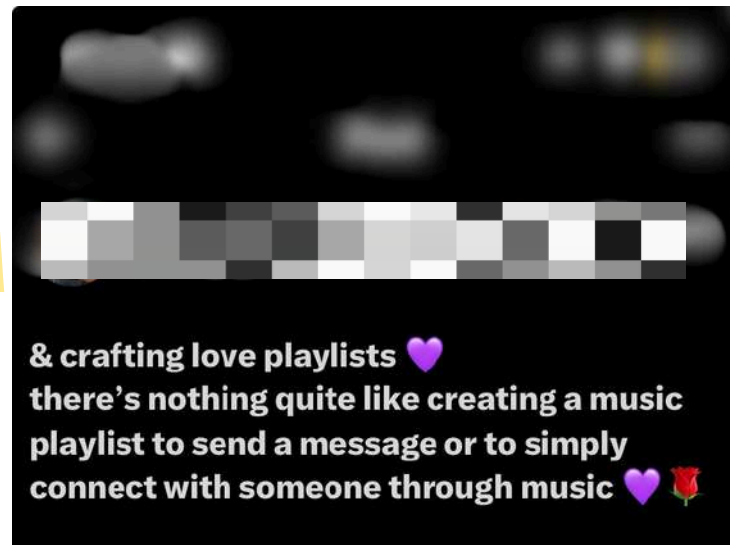
Music reflection is happening - just not in one place



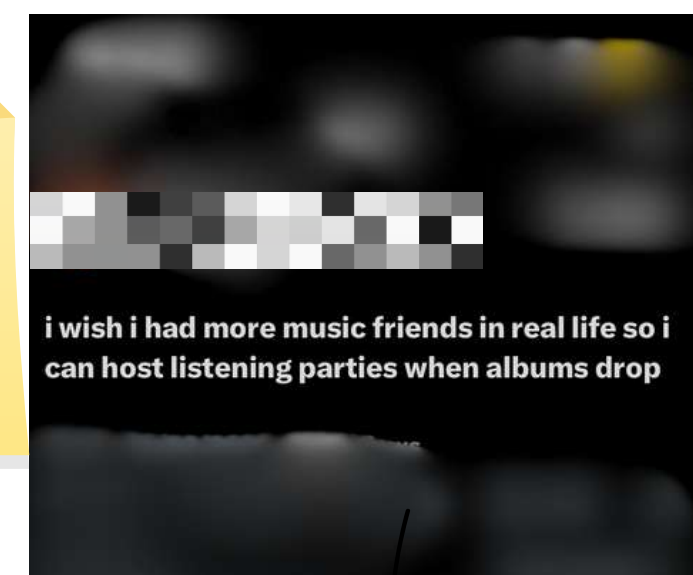
Six public X posts surfaced four ways users improvise on platforms not designed for reflection



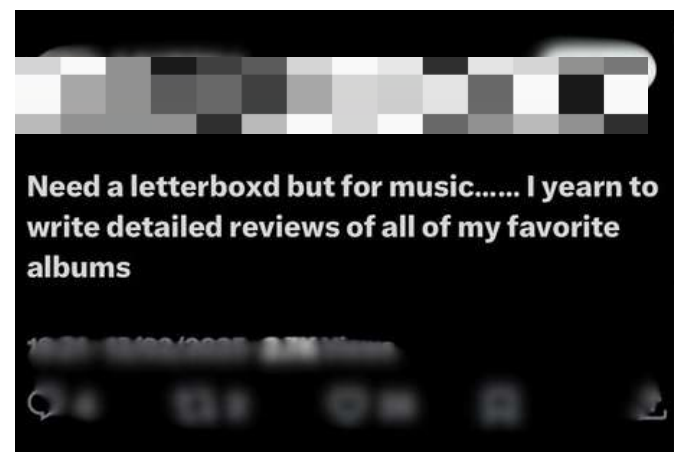
Users improvise structured listening on platforms with no shared-listening primitive



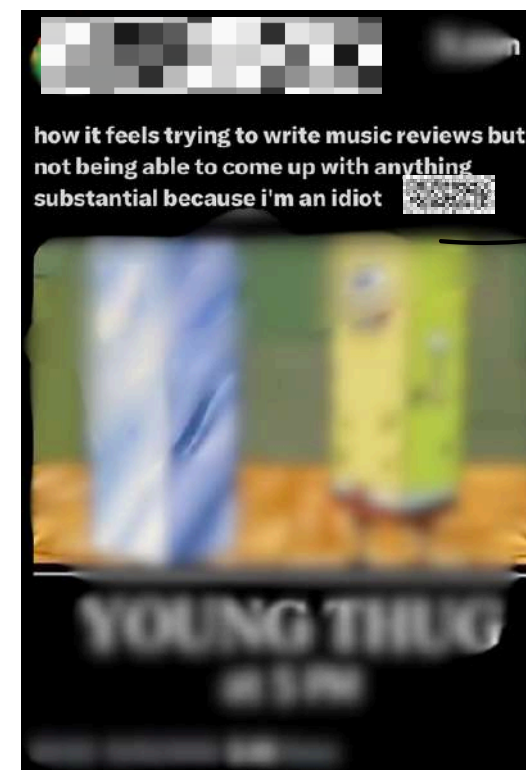
Users use playlists as a form of emotional expression and communication, but existing platforms do not support meaningful contextual sharing or deeper conversation around them.



While music is social, digital platforms do not effectively foster sustained connections around shared music interests. Users struggle to find like-minded listeners and host collective listening experiences. Additionally, feeds are cluttered with unrelated content (e.g., ads and political promotions), which disrupt music-focused engagement.



Long-form music writing happens on Substack, X, Notes, Medium, Reddit - none designated to just music



User struggles to articulate structured and meaningful music reviews due to a lack of guided writing support in existing platforms.

Synthesis: music reflection isn't missing from the internet - it's scattered across platforms not designed to host it. PeerPods is the dedicated companion layer for what users are already trying to do

What is PeerPods?



PeerPods is a context-aware augmented interface in the music and digital entertainment domain combined with a recommender system - used as a calm companion layer alongside existing music platforms for writing, community, discussion, and discovery.

It supports:

- Reflective writing — reviews, theories, thoughts, with adaptive prompts that aid structure
- Guided participation — clubs and listening parties around shared interests, not algorithmic feeds
- Meaningful peer interaction — playlist sharing with attached context, not just files
- User-controlled discovery — onboarding-driven recommendations with visible, adjustable signals



INTELLIGENCE

What makes PeerPods intelligent?

The interface adapts to the user - not the other way around



01 Adaptive prompts

Smart Prompts evolve with your draft, your interaction history, and what you are listening to – including a fun & gentle prompt when active listening is detected (dismissable)

What's intelligent: Context-aware writing scaffolding driven by user signals.

02 Adaptive content visibility

Your feed re-weights based on temporal context – release cycles, drops timing, trending discussion in your pods – and on patterns it learns from your behaviour over time – what you read deeply, return to, or, scroll past

What's intelligent: Dynamic prioritisation that responds to context, not just clicks

03 Recommender with explicit signals

Recommendations start from your onboarding signals and learn over time — observing what you read deeply, engage with, and dismiss. Selective interest-overlap from similar users supplements; every rec shows why.

What's intelligent: Cold-start by explicit signal; refines through implicit behaviour over time

04 User control as a first-class feature

Every adaptive behaviour has a visible toggle. 'Not for me' updates the model immediately. Settings expose all signals

What's intelligent: Human-in-the-loop, with control surfaced - not buried

RESEARCH APPROACH

How I gathered user evidence



01

6

X posts coded

Social Listening on X

Collected six public X posts where users articulated frustrations with the current music platforms. Each post was annotated with an insight card.

02

6

Platforms reviewed

Competitive analysis

Reviewed Letterboxd, Medium, Substack, Reddit, X, Spotify's Social Layer and Soundcloud Social Layer - the platforms users currently improvise on - to understand the gaps.

03

3

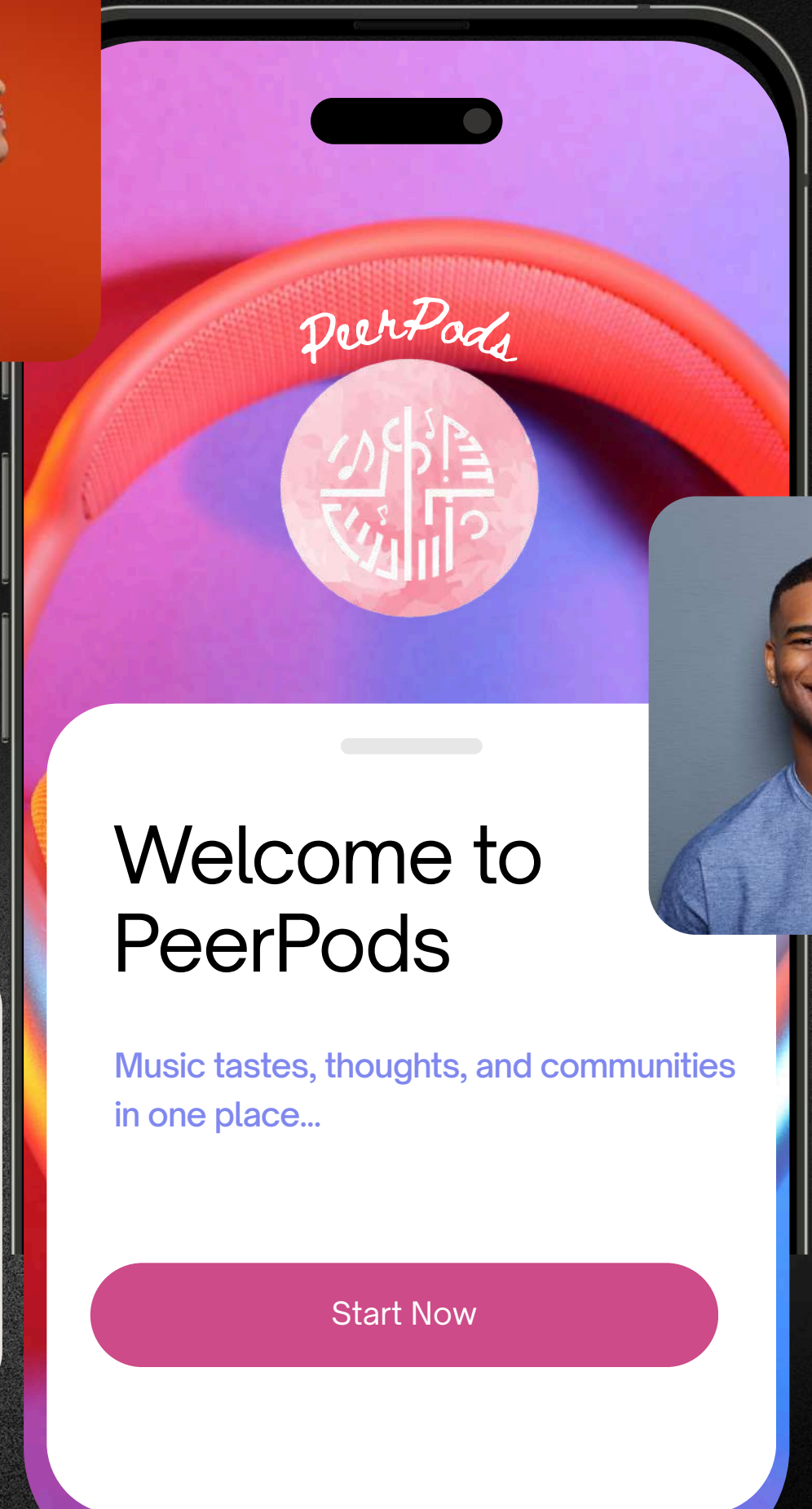
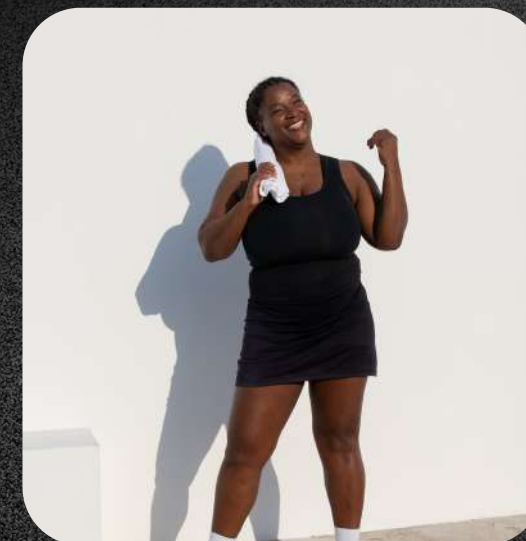
Personas synthesised

Persona Development

Synthesised three personas representing the user archetypes that emerged across both research strands.

User Persona

PeerPods



Amara Johnson

Persona 1

Age

23
Years Old

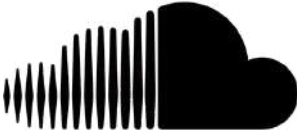
Occupation

University student
(Creative Arts)

Location

London, UK

Music apps



Gender

Female

Hobbies

Journaling, attending concerts,
discovering new music, writing
reviews

Publishing Platforms

Medium



reddit



substack

Music Behaviour

- Listens to music daily across multiple platforms
- Sometimes writes thoughts about music
- Strong interest in guided prompts for discussions
- Finds current music apps not beginner-friendly and recommendation-heavy

Needs & goals

- A dedicated space to write reflections about music
- Context-aware prompts to structure her thoughts
- Communities based on interests, but with personal control

Pain points

- Finds music apps overwhelming and cluttered
- No clear place to document thoughts about music
- Music discussions feel scattered across platforms



“I don’t just want to listen to music — I want to think about it and talk about it properly.”

Daniel Ekong

Persona 2

Age
25
Years Old

Occupation
Technical Support

Location
**Port Harcourt,
Nigeria**

Music apps



Gender

Male

Hobbies

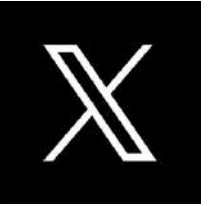
Machine Learning, Swimming,
Music, Video Essays

Publishing Platforms

ResearchGate

Medium

substack



“Sharing music
should feel more like
a conversation.”

Music Behaviour

- Listens 1–3 hours daily
- Rarely writes about music, but actively shares playlists
- Very interested in messaging and social interaction
- Finds posts get “lost in the feed” on current platforms

Needs & Goals

- Easy ways to share playlists with context
- Messaging built into the music discussion space
- Prompts that help him start conversations

Pain points

- Playlists and opinions quickly get lost in fast-moving feeds.
- Discussing music requires switching between multiple apps.
- Sharing music often lacks context and meaningful conversation.

Munachi Umeh



Persona 3

Age

35

Years Old

Occupation

Product Designer

Location

Lagos, Nigeria

Music apps



Gender

Female

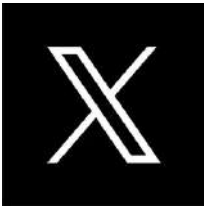
Hobbies

Listening to music,
Fixing puzzles, Plays Basketball

Publishing Platforms

Medium

substack



Music Behaviour

- Listens to music more than 5 hours daily
- Often writes and reflects on music
- Highly aware of algorithm quality and AI behaviour
- Values safe interaction and prompt-based activity

Needs & goals

- Wants well-curated, interest-based communities for deeper music discussion.
- Needs transparent, diverse recommendations beyond mainstream trends.
- Seeks control over personalisation to build trust in AI features.

Pain points

- Existing platforms prioritise trends over thoughtful music discussion.
- Music communities feel shallow and poorly curated.
- Deep conversations about music culture are hard to sustain.
- Recommendation systems feel opaque and repetitive.
- Lack of user control reduces trust in AI-driven features.



“Good design makes me trust the intelligence behind the app.”

REQUIREMENTS

From evidence to requirements

Traceability map — every requirement traces back to a user statement

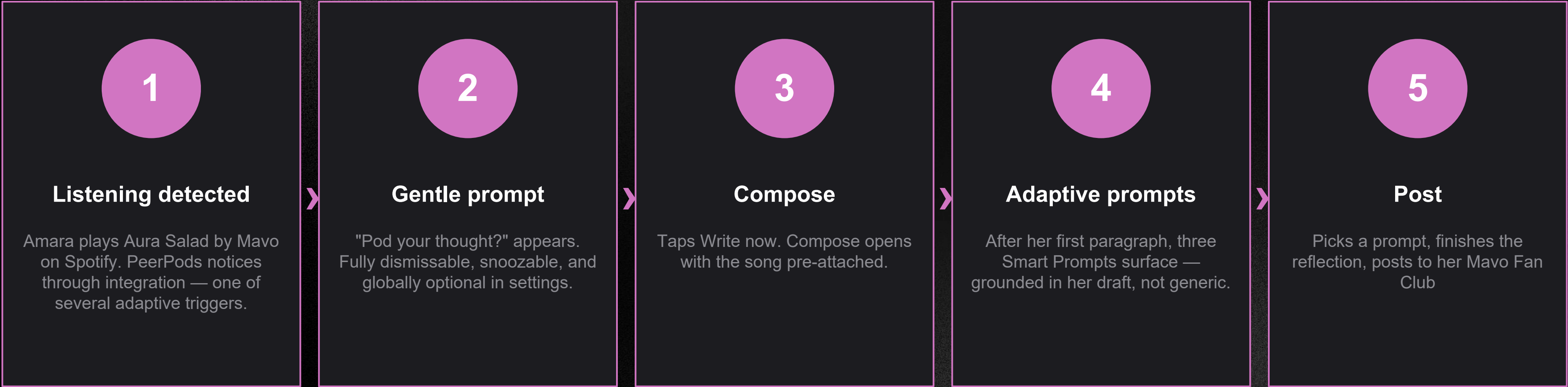


USER EVIDENCE (X)		USER NEED		FUNCTIONAL REQUIREMENT
"book club... for albums"	➔	Structured shared listening with discussion	➔	F4: Music clubs (online or in person)
"host listening parties when albums drop"	➔	Real-time community around drops	➔	F3: Album listening parties with invitations
"Need a letterboxd... write detailed reviews"	➔	Long-form structured review space	➔	F1: Structured reflections, reviews, theories
"can't come up with anything substantial"	➔	Guided writing scaffolding	➔	F2: Context-aware writing prompts
"crafting love playlists... message"	➔	Playlists with attached context	➔	F5: Playlist sharing with context
"feeds cluttered with unrelated content"	➔	Personalisation users control & trust	➔	F6: Onboarding-driven For You + opt-out

SCENARIO · PRIMARY

Amara: from listening to writing

How adaptive prompts scaffold reflection - with listening detection as one trigger



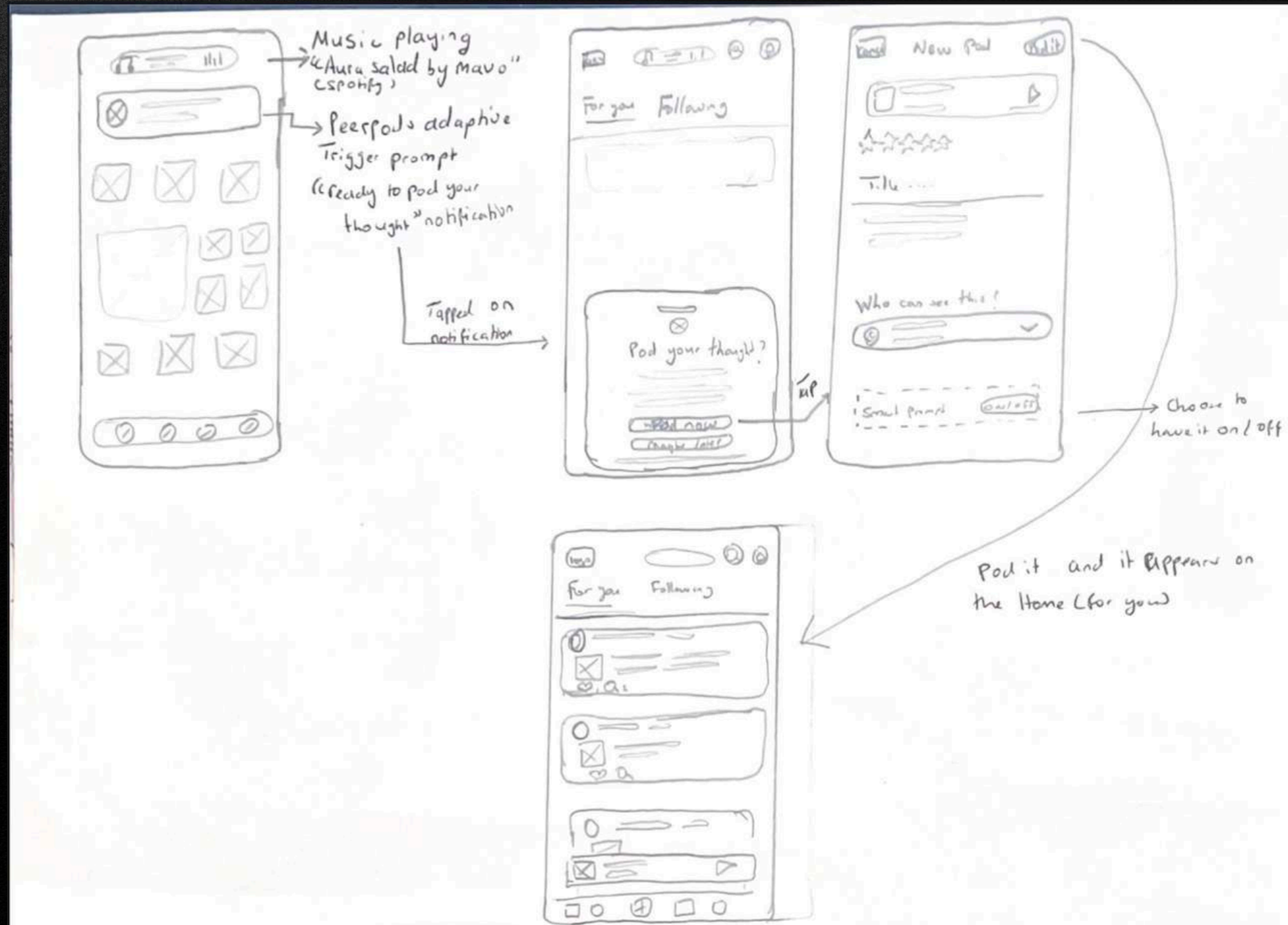
Demonstrates: F1, F2, F4 · **Intelligence features** 1, 4 (adaptive prompts + user control) · *Amershi G3, G4, G17*

SCENARIO · PRIMARY

Amara: from listening to writing



Wireframe of Scenario (Low-fidelity)



Demonstrates: F1, F2, F4 · **Intelligence features** 1, 4 (adaptive prompts + user control) · **Amershi** G3, G4, G17

SCENARIO · BREADTH

How the design serves diverse user needs



Daniel: a drop, surfaced and shared

Adaptive content visibility around release cycles

- 1 Feed re-weights as Tyler, the Creator's release window approaches
- 2 Drop surfaces with a 'Listen together?' option in his Tenski Club (Alte)
- 3 He invites the club; 14 join a synced session with chat
- 4 Posts a quick reflection — adaptive prompts help him start

Demonstrates: F3, F4, F6 · Intelligence 2, 1 (adaptive visibility + prompts)

Munachi: correcting her feed

User control as a first-class behaviour

- 1 Long-presses a mainstream pop rec → 'Not for me'
- 2 Selects a reason: 'I want more underground'
- 3 Opens Settings → Personalisation, sees feed signals
- 4 Adjusts sliders; feed re-reasons immediately, with confirmation

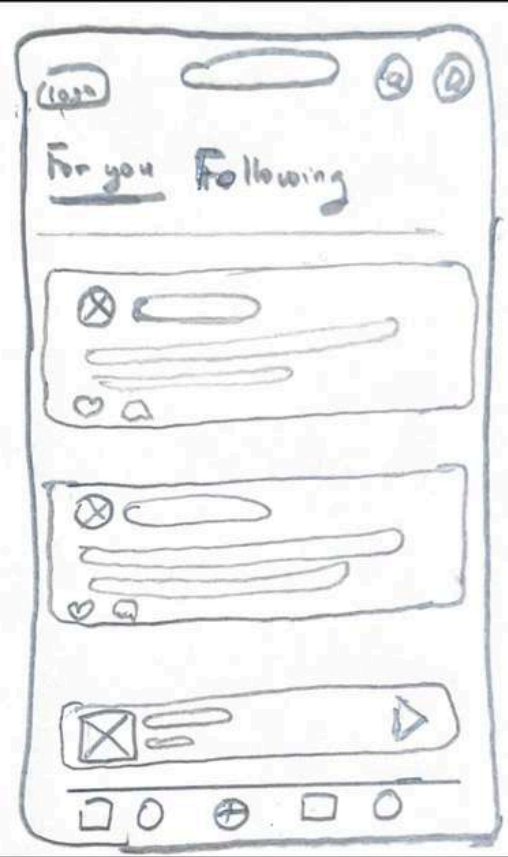
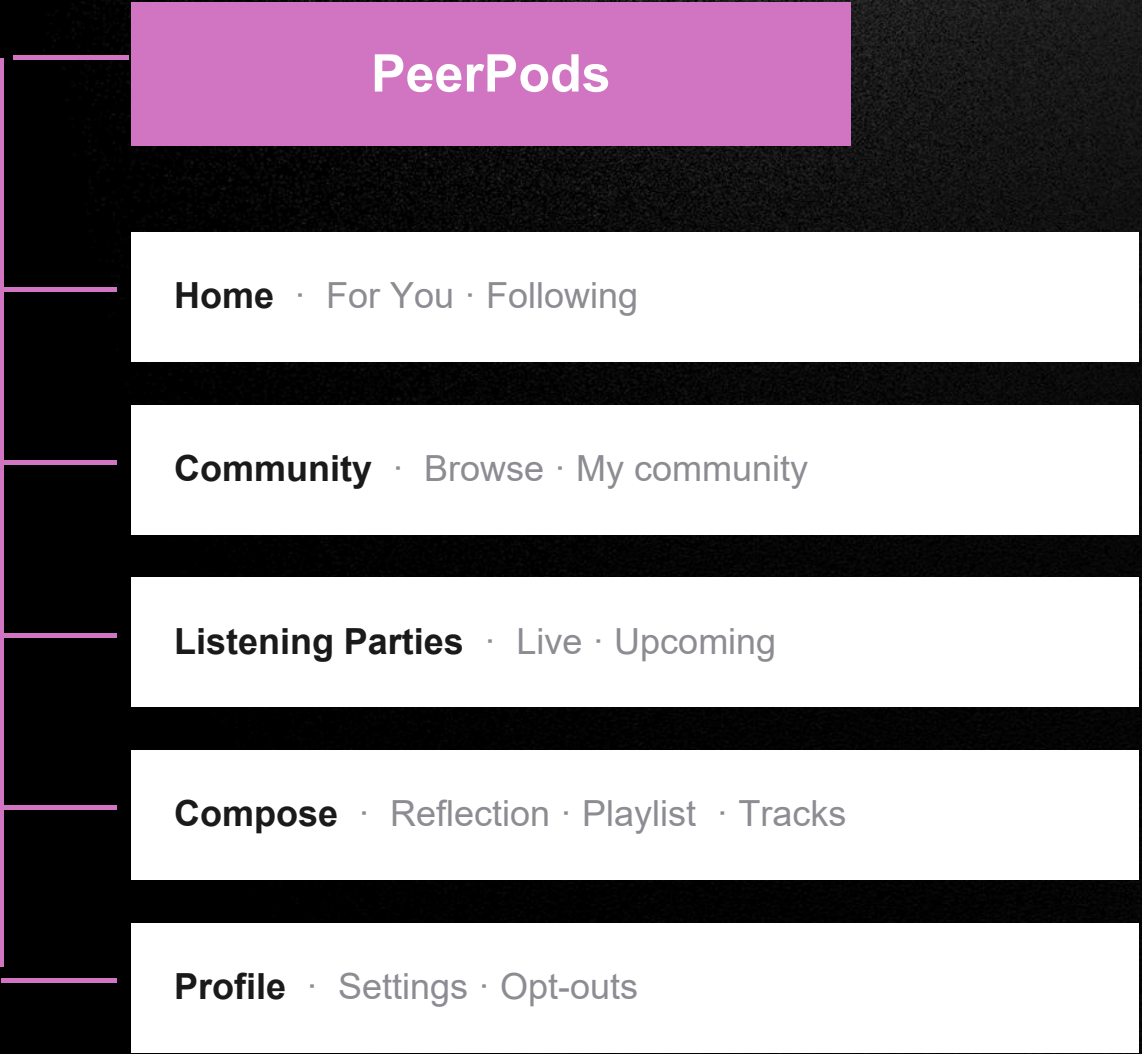
Demonstrates: F6 · Intelligence 3, 4 (recommender + user control)

INFORMATION ARCHITECTURE

FROM IA TO WIREFRAMES



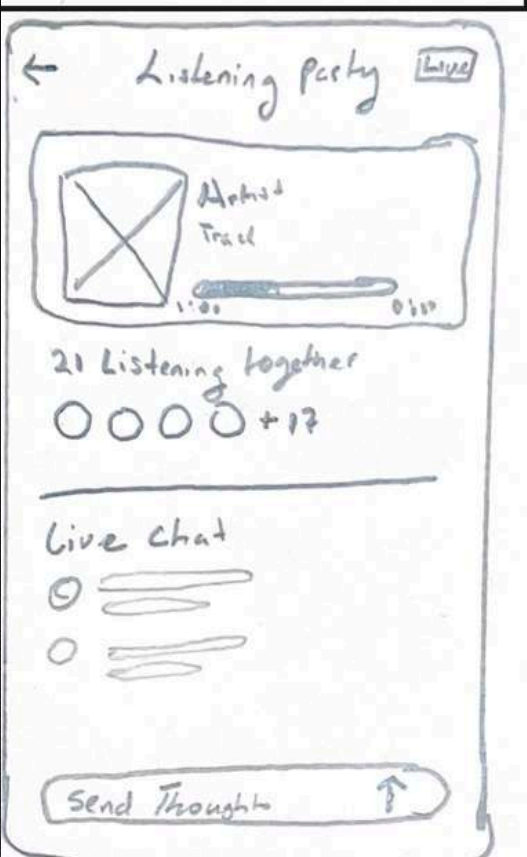
Sitemap



Home (For You)



Community



Listening Party



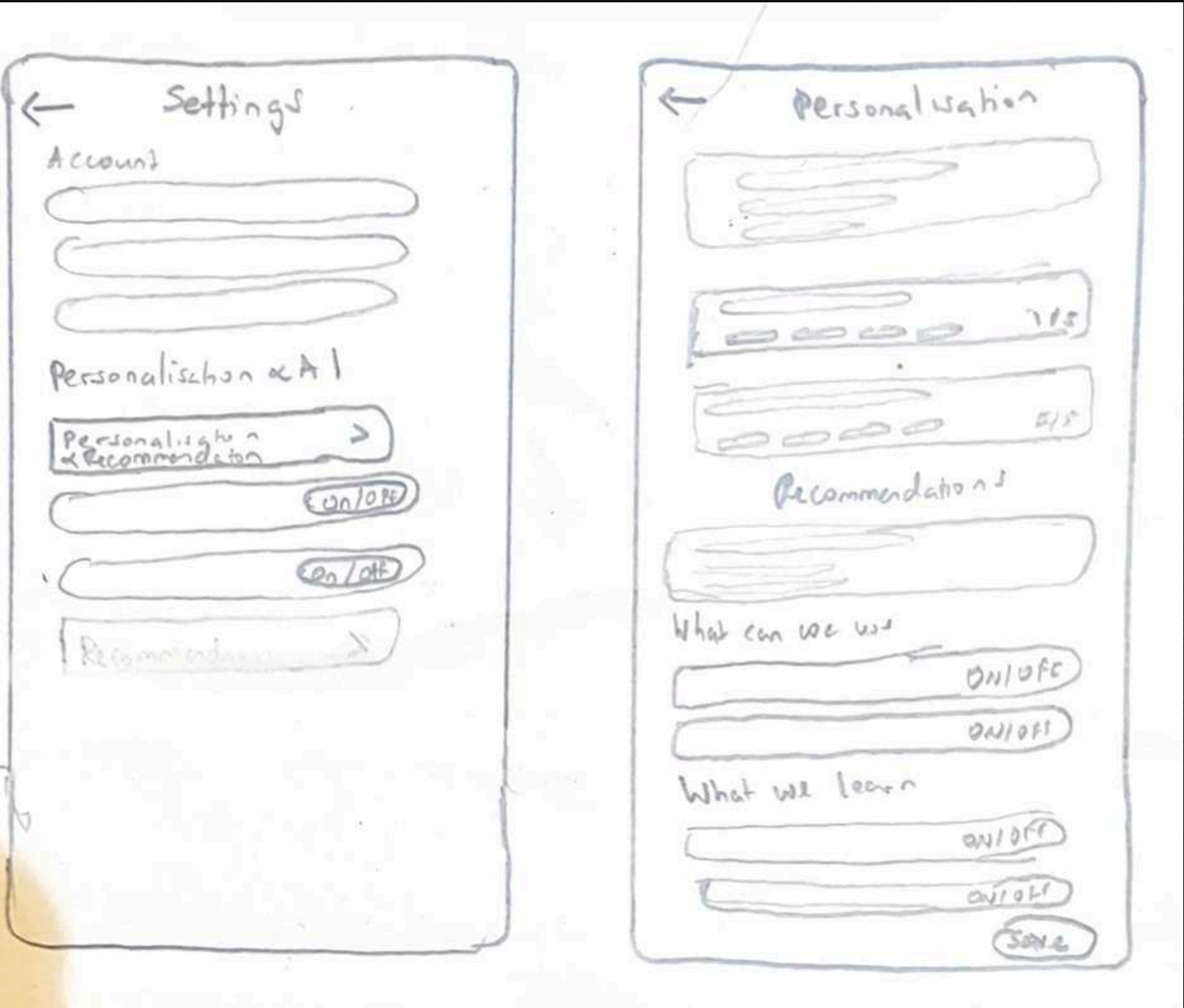
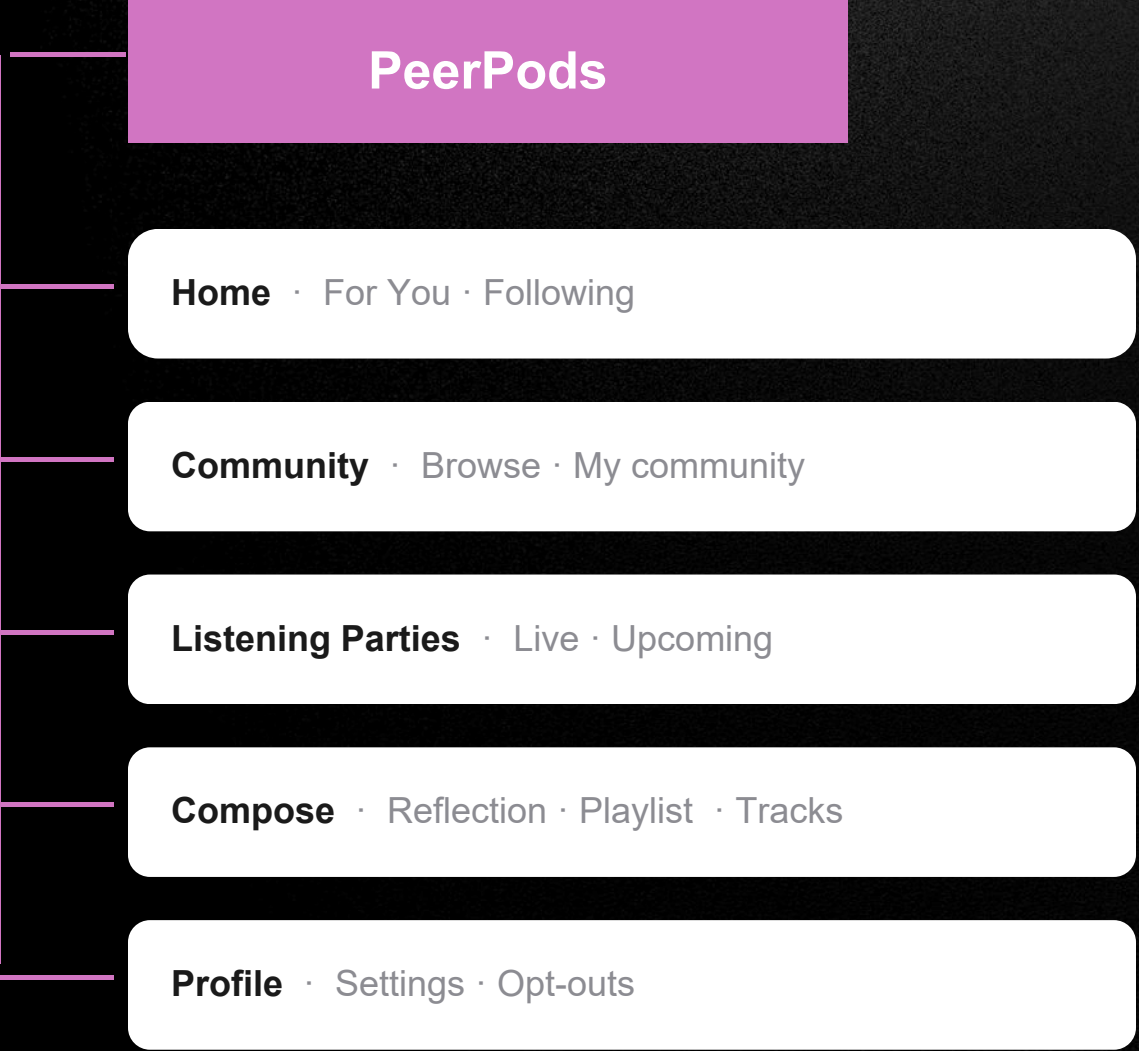
Compose

INFORMATION ARCHITECTURE CONT'

FROM IA TO WIREFRAMES



Sitemap



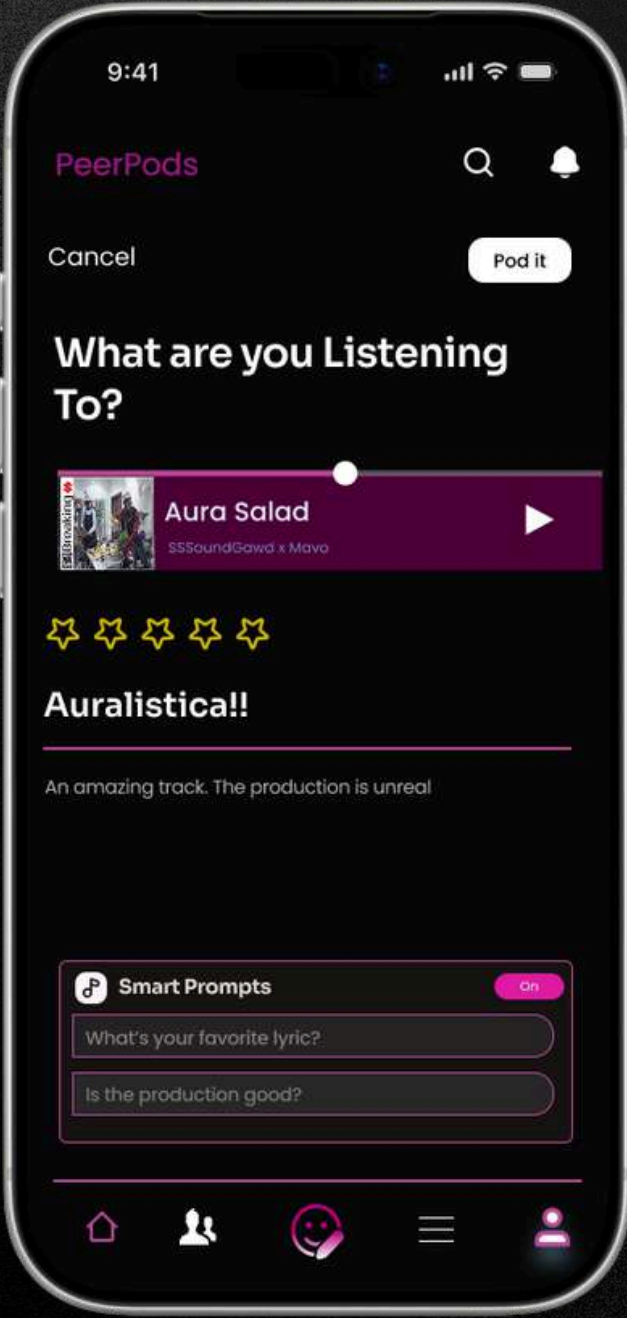
Profile - Settings - Opts-outs

PROTOTYPE · THE PRODUCT

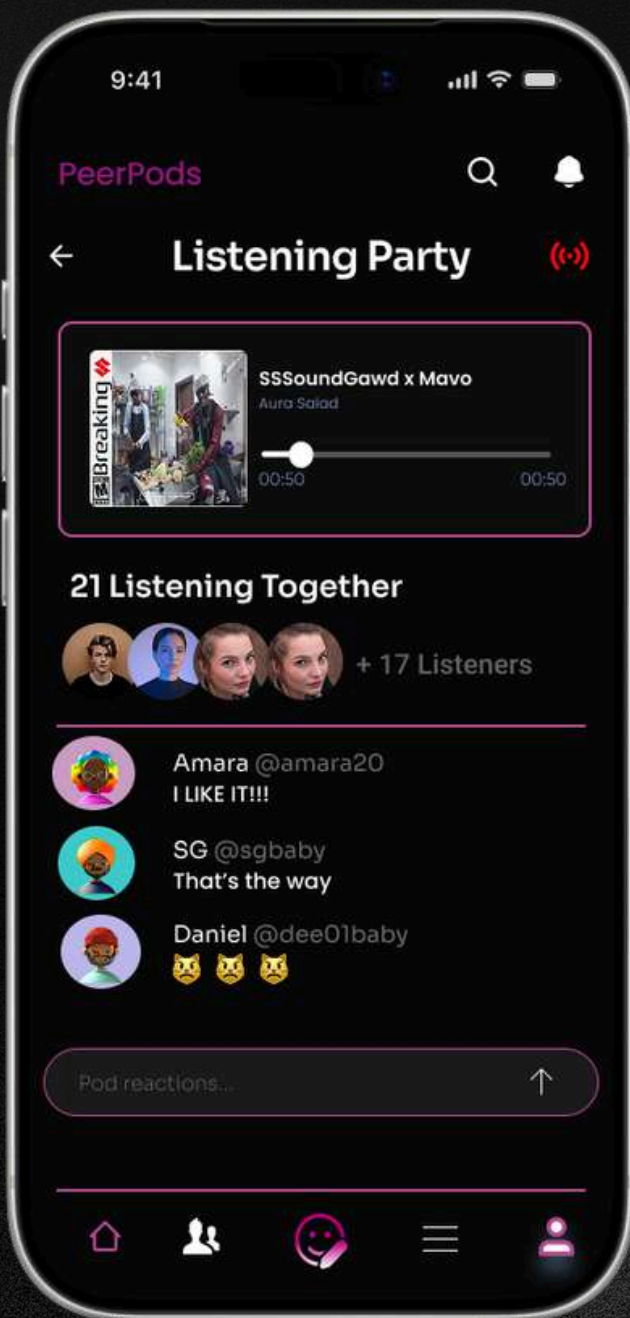
PEERPODS — HIGH-FIDELITY PROTOTYPE



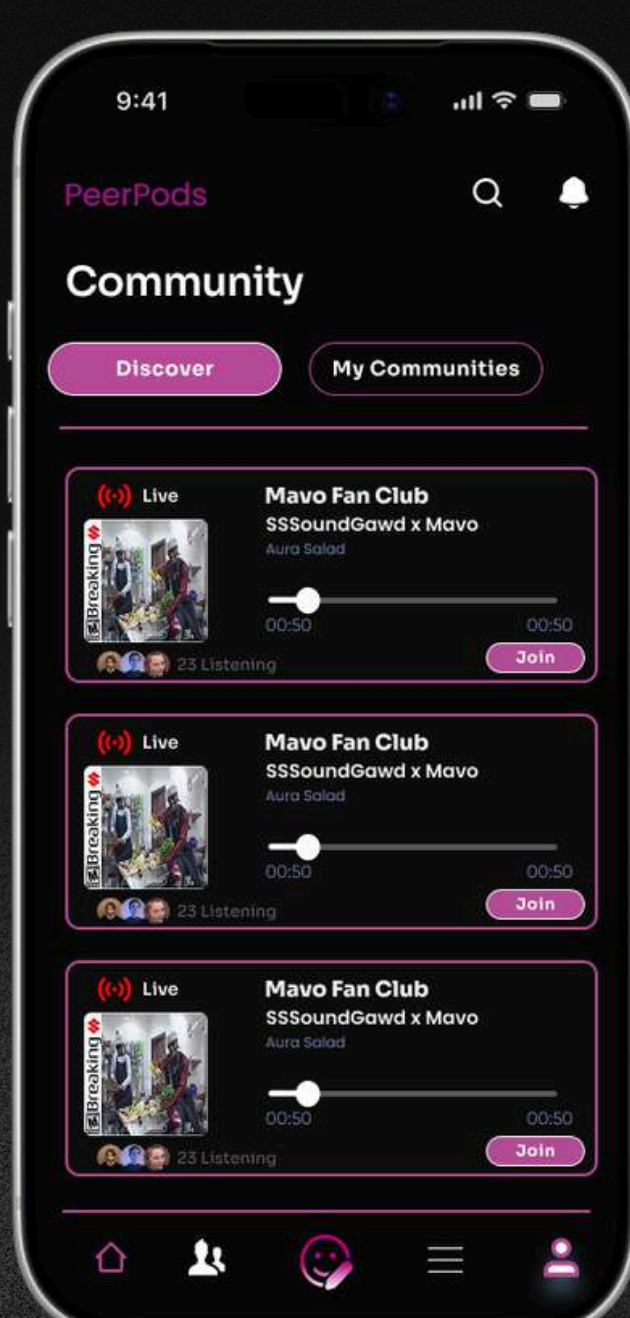
Home — For You



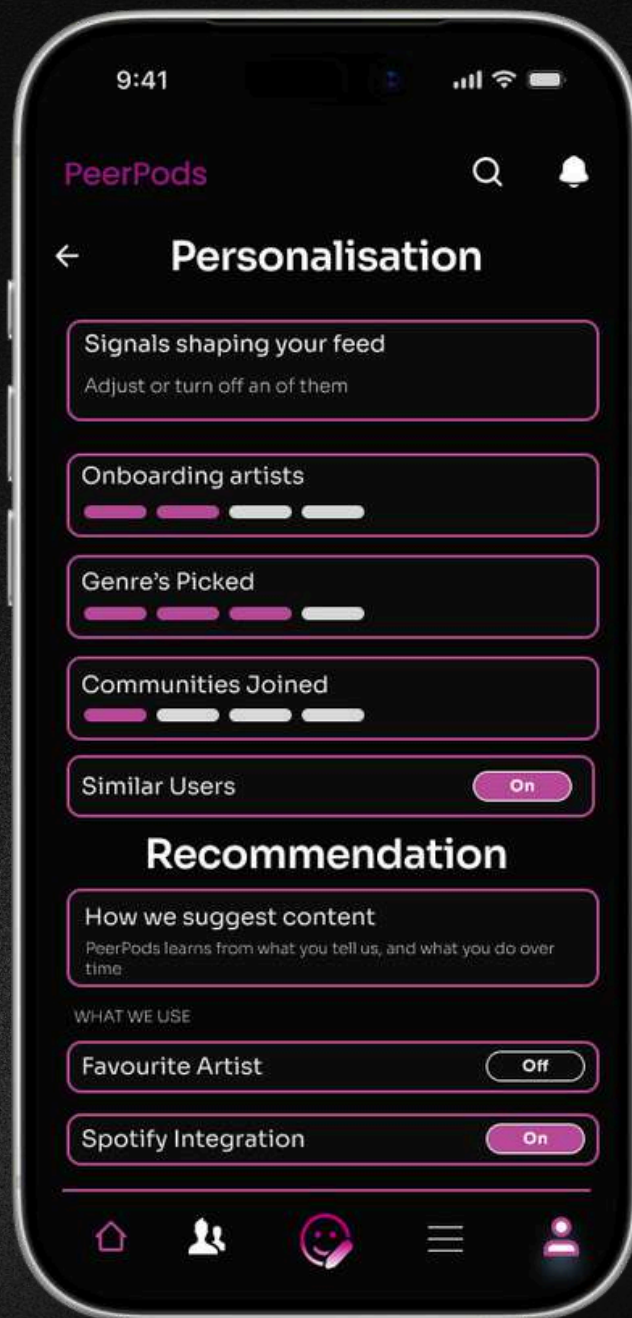
Compose



Listening Party



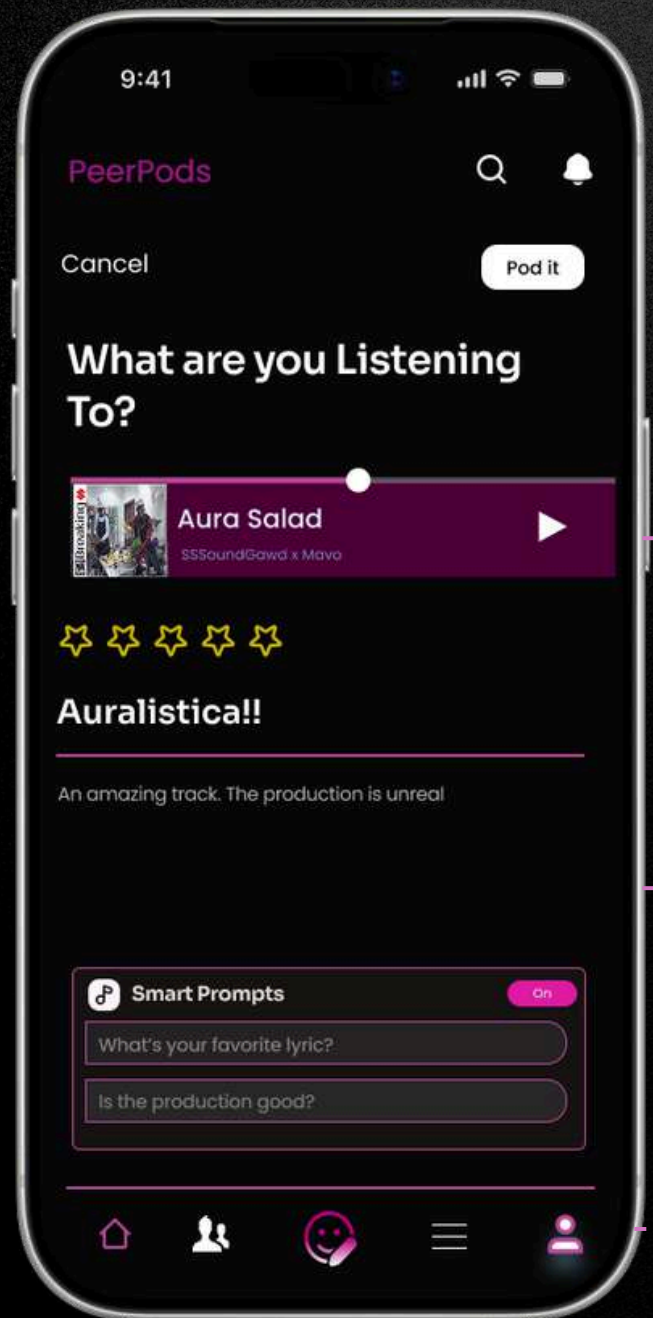
Community



Personalisation & Recommendation

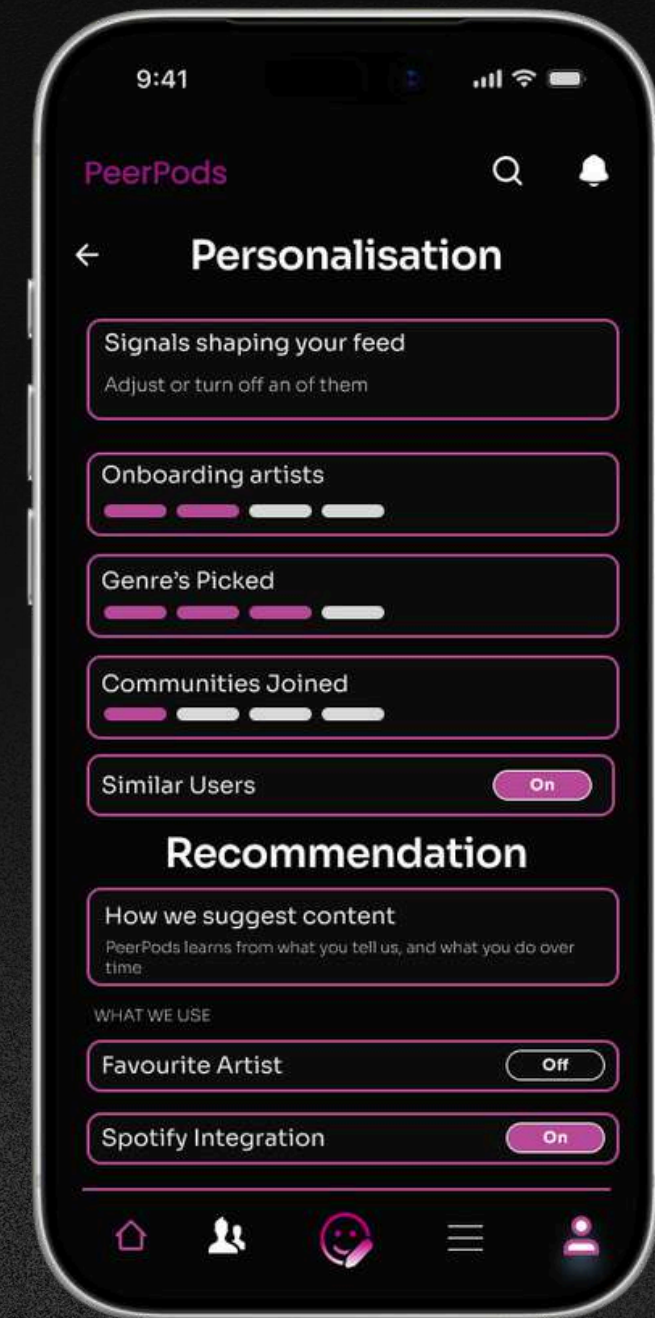
PROTOTYPE · THE INTELLIGENCE

WHERE THE INTELLIGENCE IS VISIBLE

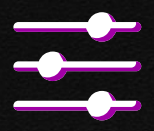
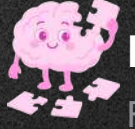


Compose

-  **Listening detected**
Track auto-attached
Amershi G4: contextual relevance
-  **Smart Prompts drawer**
Surfaces after 1st paragraph; toggle visible
Amershi G1 + G3: capabilities + timing
-  **Off switch**
Global control in settings
Amershi G17: global controls
-  **Adaptive over time**
Prompts refine from your draft history
Implicit learning from behaviour



Personalisation & Recommendation

-  **Signal sliders**
User can see and adjust feed inputs
Amershi G11: show why
-  **Not for me**
Granular dismiss with reason capture
Amershi G15: granular feedback
-  **Live Updates**
Feed re-reasons immediately on change
Amershi G16: incorporate user input
-  **Learns over time**
Reading patterns refine from behaviour
Implicit signals as recommender input

EVALUATION

THINK-ALoud PROTOCOL WITH USERS



PROTOCOL & TRIGGER

Setup: Teams call, screen-share, audio + screen recorded; participant told to verbalise thoughts continuously.

Five triggers to keep users talking:

- **Help** — if frustrated, encourage and steer back
- **Reminder** — if silent for >10s, ask them to keep talking
- **Echoing** — repeat partial statements to invite finishing
- **Ask explanation** — on irritation, ask why
- **Ask suggestion** — on dislike, ask what would work better

EVALUATION TASKS

TID	Task	Purpose	User
T01	Signup & Onboarding	Cold-start signal capture	All
T02	Write a reflection on a song	Smart Prompts utility	Amara
T03	Find & join a music community	Pod discovery + transparency	All
T04	Host a listening party for a drop	Real-time shared listening utility	Daniel
T05	Dismiss a rec & explain why	Granular dismiss + reason capture	Munachi
T06	Adjust feed signals	Personalisation panel utility	Munachi

FINDINGS

WHAT THE PILOT SURFACED

Findings split by usability (can they navigate?) and utility (can they do what they came to do?)



USABILITY · FRICTION → REDESIGN

T02 — Smart Prompts timing:

Amara: "why is it interrupting me before I've even started?"

→ Gated prompts to surface only after 1st paragraph

T05 — Dismiss action discoverability:

Munachi: "how do I get rid of this one?"

→ Added long-press affordance hint on first dismiss

T04 — Listening party invite flow:

Daniel: "I expected to invite the community in one tap"

→ Reduced invite from 3 steps to 1; community pre-selected

UTILITY · IS THE SYSTEM FIT FOR PURPOSE?

T02 — Smart Prompts genuinely scaffold writing:

Amara: "oh, that's actually a good question — okay let me try"

→ Utility confirmed for the writing-with-prompts flow

T06 — Personalisation panel earns trust:

Munachi: "this is what I want from every app actually"

→ Utility confirmed; transparency is the differentiator

T03 — Pod discovery: utility gap surfaced:

Daniel: "I want to see what they're actually talking about first"

→ Added recent-discussion preview on pod cards

REFLECTION

On process and final design

Three honest learnings from running this UCD process end-to-end



01

Where the design genuinely helps

- Amara's full scenario showed how the system recognised context, scaffolding writing and giving the option opt-out - three intelligence behaviours.
- Munachi's experience transparency; she was able to adjust the signals, which curated her feed.
- The recommender enhances the accuracy of the behaviour over time, not just explicit input; this closes the gap between user intent and feed quality

02

Where it falls short

- Cold-start improves with onboarding-driven signals but still gives weaker recs in the first week
- The writing-first framing may exclude users who reflect through voice, photo, or video — a real accessibility consideration
- Three pilot participants surface friction but can't measure long-term engagement; deeper longitudinal study is needed

03

Non-trivial process learnings

- Traceability in design necessitated correlating every requirement with a tweet, ensuring that only substantiated features were included.
- The think-aloud method provided valuable insights that might have been overlooked in a traditional evaluation, highlighted by comments reflecting genuine engagement.
- Additionally, personas serve best as decision-making tools throughout the process instead of being treated merely as artifacts.

REFERENCES

- Amershi, S., Weld, D., Vorvoreanu, M., Fournery, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P.N., Inkpen, K., Teevan, J., Kikin-Gil, R. and Horvitz, E. (2019) 'Guidelines for Human-AI Interaction', in Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. Glasgow, 4–9 May. New York: ACM, pp. 1–13.
- Höök, K. (2000) 'Steps to take before intelligent user interfaces become real', *Interacting with Computers*, 12(4), pp. 409–426.
- ISO (2019) ISO 9241-210:2019 Ergonomics of human-system interaction — Part 210: Human-centred design for interactive systems. Geneva: International Organization for Standardization.
- Liao, Q.V. and Vaughan, J.W. (2024) 'AI Transparency in the Age of LLMs: A Human-Centered Research Roadmap', *Harvard Data Science Review*, Special Issue 5.
- Nielsen, J. (1994) *Usability Engineering*. San Diego: Morgan Kaufmann.
- Norman, D. (2013) *The Design of Everyday Things*. Revised edn. New York: Basic Books.
- Shneiderman, B., Plaisant, C., Cohen, M., Jacobs, S., Elmqvist, N. and Diakopoulos, N. (2017) *Designing the User Interface: Strategies for Effective Human-Computer Interaction*. 6th edn. Boston: Pearson.

